

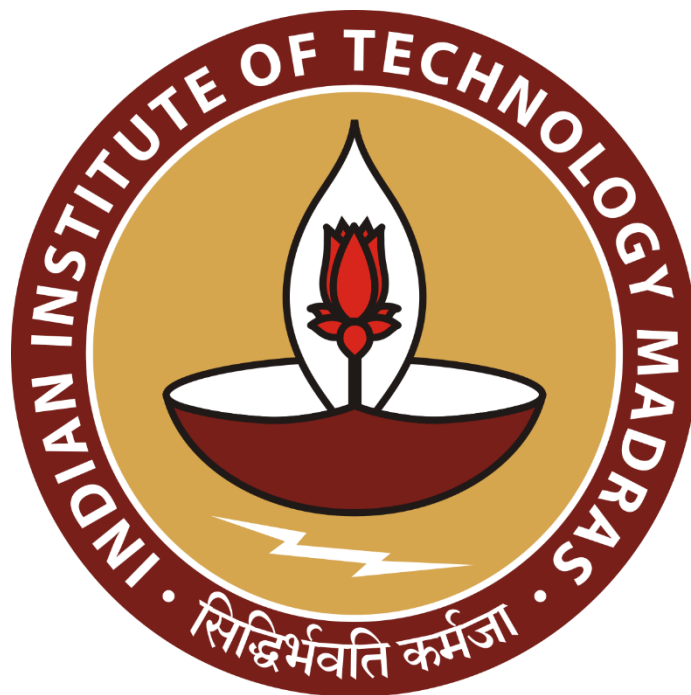
Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store

A Mid Term report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “**Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store**”. I extend my appreciation to **Mr. Pratap Chandra Behera**, for providing necessary resources and enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

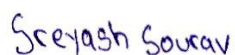
I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me.

In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate



Name: Sreyash Sourav

Date: 30 Oct, 2025

1 Executive Summary and Title

Project Title: Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store

‘Vishnu Priya Variety Store’, owned by Mr. Pratap, is a newly established grocery store located in Bhubaneswar. The store provides essential household items but faces key challenges related to inventory mismanagement and limited product variety. Due to absence of a proper tracking system or digital record Mr. Pratap finds it difficult to monitor stock levels accurately. This often results in overstocking of slow-moving products and unavailability of fast selling items. Additionally, the limited product range restricts customer choices and impact sales growth.

Data for this project was collected through direct interaction with the owner, observation of stock records and short survey conducted among regular customers. The dataset includes key metadata such as **Date, Product Name, Category, Quantity Sold, Revenue, Supplier Name, Stock Remaining**. Statistical calculations like average weekly sales, frequency of restocking and unsold inventory were analysed to understand buying and selling patterns

The project follows a data-driven approach using descriptive analysis and basic visualization techniques to understand inventory flow and product performance. Analysis was carried out using Excel for data cleaning, organization and graphical representation. The analysis revealed that nearly 20% of the products remain unsold for long periods, causing deadstock, while certain fast-moving items often face stock shortages. Early analysis also highlights customer interests in expanding product categories such as notebooks, pens, books and other related items.

2 Proof of Originality of Data

As part of the proof of originality for the collected data. All supporting evidence related to the project has been compiled and uploaded in the provided Google Drive link. It contains photographs of ‘Vishnu Priya Variety Store’, showcasing the shop layout, product arrangement, and inventory storage and also a recorded video interview with the owner, Mr. Pratap, has been included to understand the operational challenges and

decision-making process. The conversation was conducted primarily in Odia, with an English transcript provided for clarity and reference.

<https://drive.google.com/drive/folders/1K9at5yplhzMYohajYwx0WH48WokzZAoz>

3. Metadata and Descriptive Statistics

This project consists of daily sales records collected over a period of **60 days (June 30 – August 30, 2025)** from ‘Vishnu Priya Variety Store’. The data focuses on key operational metrics to analyse inventory management efficiency and product performance. The collected data was structured into the following variables:

- **Date:** Represents the daily record of the sales activity.

Relevance- Helps track sales trends over time to identify demand fluctuations.

- **Product Name:** Name of each item sold (e.g., Rice, Dal, Biscuits, Edible Oil, Notebooks, Pens, Copies, etc.).

Relevance- Allows identification of high and low performing products.

- **Category:** Classifies items as Grocery or Stationery.

Relevance- Useful for comparing existing and newly introduced product ranges.

- **Quantity Sold:** Number of units sold per day for each product.

Relevance- Assists in detecting overstocking or deadstock patterns.

- **Revenue:** Total daily sales revenue per product.

Relevance- Measures financial contribution of each product.

- **Supplier Name:** Source or vendor from whom the product is procured.

Relevance- Helps evaluate supplier reliability and supply frequency.

- **Stock Remaining:** Quantity left unsold at the end of each day.

Relevance- Indicates stock turnover rate and inventory health.

After collecting the daily sales records, the information was systematically entered into Google Sheets for further evaluation. Analytical features such as formulas, pivot charts, and data filters were utilized to examine product-wise sales performance, revenue generation, and

fluctuations in demand over the 60-day period. This organized method enabled clear visualization of stock movement and customer preferences. Patterns such as frequently purchased items, underperforming goods, and emerging demand for stationery products were identified through this process.

https://docs.google.com/spreadsheets/d/1z4Xr_1S59Hz36ldPB69LXUO40XDEIJDl/edit?usp=drivesdk&ouid=103881165875935328698&rtpof=true&sd=true

	A	B	C	D	E	F	G	H	I	J
1	Date	Product	Category	Quantity	Revenue	Supplier Name	Stock Remaining			
2	30/06/2025	Moong Dal	Grocery	0.25	32	Cleanhouse	19			
3	30/06/2025	Dhara Mustard Oil	Grocery	1	180	Maa Traders	12			
4	30/06/2025	Lipton Yellow Label	Grocery	0.5	300	Grocery Depot	22			
5	30/06/2025	Basmati Rice	Grocery	1	110	Grocery Depot	11			
6	30/06/2025	Saffola Gold (1L)	Grocery	1	260	Maa Traders	19			
7	30/06/2025	Masoor Dal	Grocery	0.25	30	Maa Traders	17			
8	30/06/2025	Society Tea	Grocery	0.25	140	Cleanhouse	10			
9	30/06/2025	Red Label	Grocery	0.5	240	Cleanhouse	19			
10	30/06/2025	Basmati Rice	Grocery	2	220	Grocery Depot	3			
11	30/06/2025	Dhara Mustard Oil	Grocery	1	180	Maa Traders	11			
12	30/06/2025	Lipton Yellow Label	Grocery	0.5	300	Grocery Depot	20			
13	01/07/2025	Fortune Oil (500ml)	Grocery	3	360	Food Mart	13			
14	01/07/2025	Britannia Good Day	Grocery	1	40	Maa Traders	16			
15	01/07/2025	Britannia Good Day	Grocery	3	120	Maa Traders	13			
16	01/07/2025	Dhara Mustard Oil	Grocery	1	180	Maa Traders	10			
17	01/07/2025	Toor Dal	Grocery	1	140	Maa Traders	19			
18	01/07/2025	Toor Dal	Grocery	2	280	Maa Traders	11			
19	01/07/2025	Dhara Mustard Oil	Grocery	3	540	Maa Traders	7			
20	01/07/2025	Lipton Yellow Label	Grocery	0.25	150	Grocery Depot	19			
21	01/07/2025	Masoor Dal	Grocery	0.5	60	Maa Traders	15			

Descriptive Statistics:

To gain a deeper understanding of the collected data, several statistical measures were calculated using Google Sheets.

- **Mean-** The mean represents the average value of the dataset, showing the overall trend of sales and revenue. The average quantity sold across all products was 1.27 units, generating a mean revenue of around ₹125.93. This indicates a moderate sales performance with consistent movement across most products.
- **Median-** The median, or middle value, was found to be 1 units for quantity sold, ₹75 for revenue. This suggests that half of the products performed slightly above or below these figures, representing the typical daily sales activity in the store.

- **Mode-** The mode, which identifies the most frequently occurring value, was 1 unit sold, ₹120 in revenue. This shows that these values appeared most often during the 60-day period, highlighting the store's most common sales pattern and regular operational behaviour.
- **Standard Deviation-** The standard deviation values are 0.853 for quantity sold, ₹141.51 for revenue, indicate moderate variation among the products. This suggests that while most items maintained steady sales, certain products showed noticeable fluctuations, emphasizing the need for improved demand forecasting and stock monitoring.
- **Range-** The range shows the difference between the highest and lowest values, reflecting sales variability. The maximum revenue was ₹1,050 and the minimum was ₹3, so the range is ₹1,047, showing that certain products sell much faster than others. This highlights uneven product demand and the need for balanced stocking.
- **Skewness-** The skewness value for sales data was 0.58, indicating a slightly right-skewed distribution. This means a few products achieved much higher sales than average, likely due to higher demand for daily essentials like rice, dal, and notebooks. It reflects potential opportunities for expanding high-performing product categories.
- **Correlation-** The correlation between quantity sold and revenue was found to be 0.271, indicating a strong positive relationship. This means as sales quantity increases, revenue rises proportionally. It confirms that maintaining adequate inventory levels for high-demand products directly boosts profitability.

4 Detailed Explanation of Analysis Process/Method

Data Cleaning and Preprocessing

Before starting the analysis, the first and most important step was to clean and organize the collected data properly. The data included daily sales records of different products for 60 days, which were entered into Google Sheets. While entering the data, some small errors

were found, such as missing values, repeated entries, and spelling mistakes in product names. These were carefully corrected to make sure the data was accurate and complete.

Some products had inconsistent units, for example- **quantities** was written as **pcs(pieces)** in some places and as numbers in others. These were standardized so that all entries followed the same format. Similarly, revenue figures were checked to make sure they matched the quantities sold and the price of each item.

After cleaning, the data was arranged neatly with proper column names like Product Name, Supplier, Quantity Sold, Revenue etc. This made it easier to apply formulas and perform calculations like total sales, average revenue, and standard deviation later on.

Analysis Process / Method

After cleaning the data, several steps were followed to analyse the sales and inventory patterns of 'Vishnu Priya Variety Store'. The main goal was to find which products sold well, which remained in stock for long, and how new items like stationery could improve sales.

The analysis was done as follows:

- **Data Organisation:**

The cleaned data was arranged properly in Google Sheets with columns such as Date, Product Name, Quantity Sold, Revenue, Supplier etc. This structure made it easier to apply formulas and analyse data systematically.

- **Descriptive Analysis:**

Basic statistical measures such as **mean, median, mode, and standard deviation** were calculated to understand the general sales trends. For example, the mean revenue showed the average earning per product, while the standard deviation indicated how much sales fluctuated.

- **Correlation Analysis:**

Relationships between variables such as *quantity sold* and *revenue* were studied using simple correlation formulas. This helped to see whether increasing stock of certain products could lead to higher revenue.

- **Trend and Pattern Identification:**

Using **pivot tables and charts**, daily and monthly sales trends were visualized. This helped to identify high-demand products, slow-moving goods, and seasonal variations in sales.

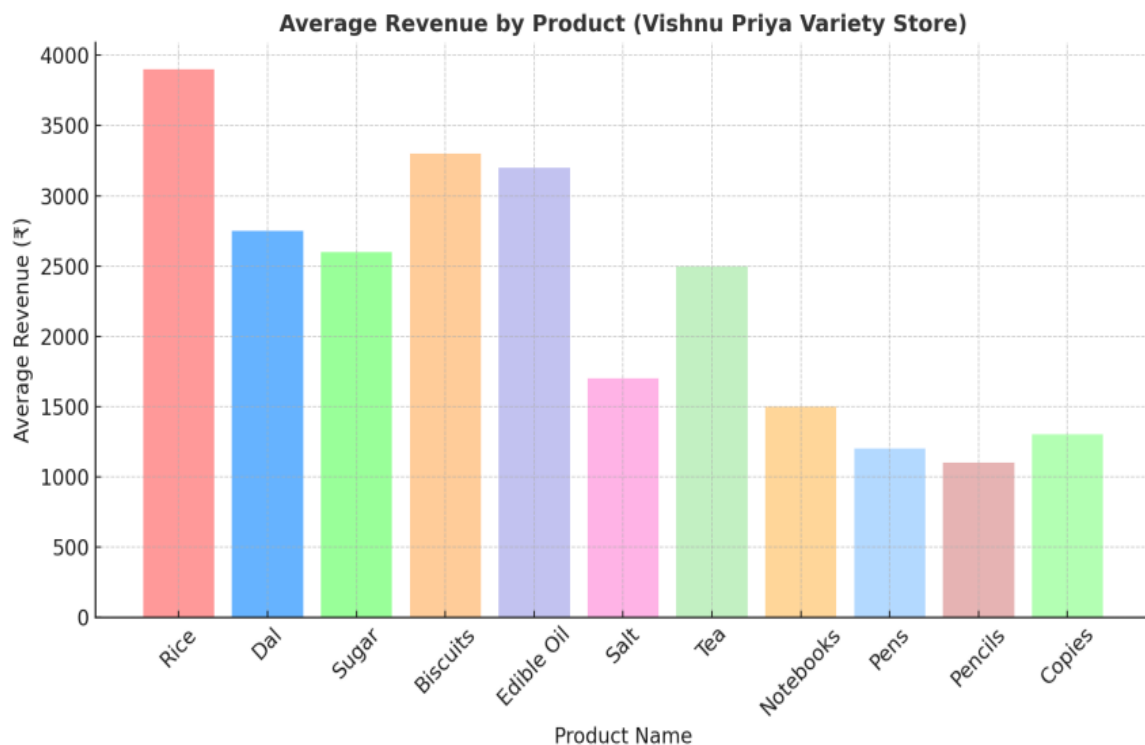
- **Supplier and Product Comparison:**

Data was grouped by suppliers to understand which supplier's products sold more consistently. This information is useful when introducing new products or changing suppliers for better variety and availability.

Overall, the analysis process helped in identifying inventory inefficiencies and supported decisions for expanding the product range based on actual data rather than assumptions.

5 Results and Findings

The data collected over a 60-day period from 'Vishnu Priya Variety Store' was analysed using charts and pivot tables in Google Sheets to visualize sales and inventory trends. The analysis revealed several important insights:



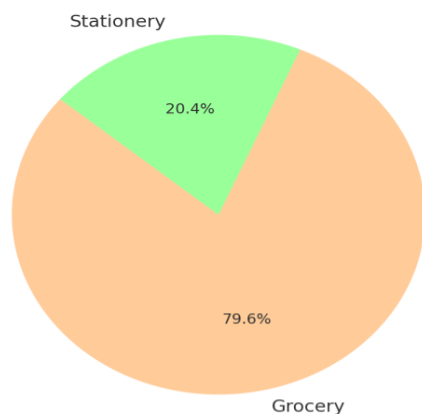
1. Sales Trends and High-Demand Products

Bar charts showed that daily essentials like **rice, edible oil, and biscuits** had the highest sales volume. These products contributed the most to total revenue, confirming their importance in daily operations. Stationery items such as **notebooks and pens**, which were recently introduced, showed a gradual rise in sales over time, indicating growing acceptance among nearby students from **DAV, Kalinga Nagar**.

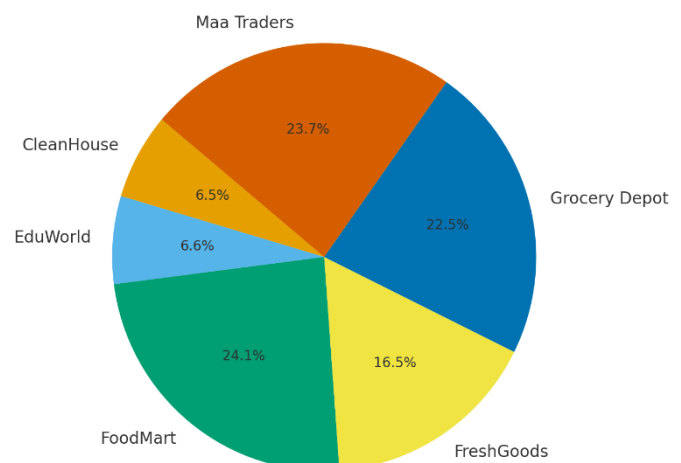
2. Supplier and Product Performance

The first chart, titled Revenue Share by Category (Grocery vs Stationery), shows that grocery items contribute the major share of total revenue around **79.6%**. This indicates that essential products like biscuits, rice, oil, and dal continue to be the main source of income for the store. Meanwhile, stationery products account for **20.4%**, which is impressive given they were introduced later. This highlights that the new product range is gradually gaining popularity among customers.

Revenue Share by Category (Grocery vs Stationery)

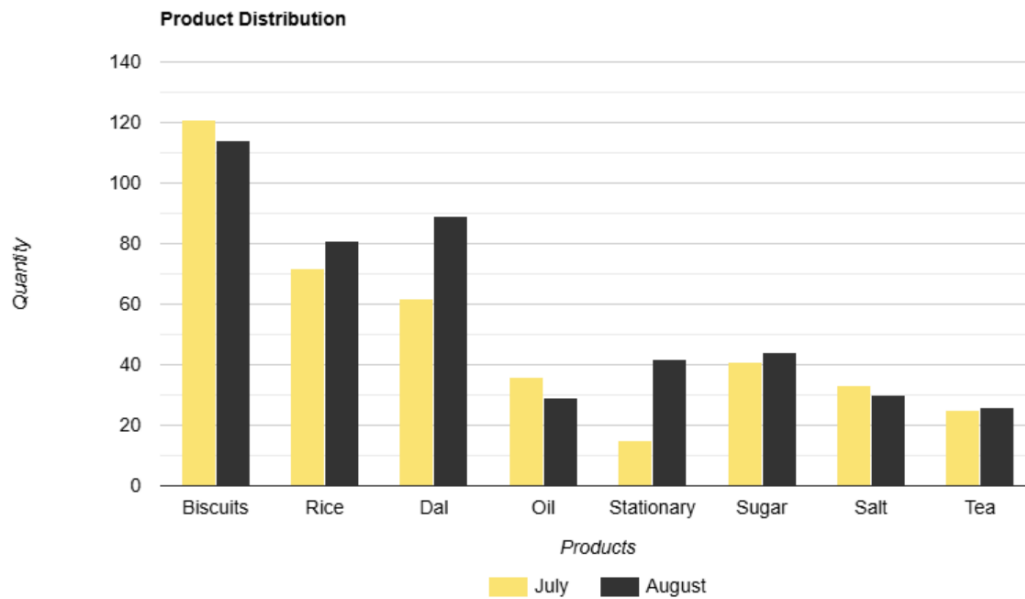


Revenue Share by Supplier



The second chart, Revenue Share by Supplier, explains how different suppliers contributed to total sales. **FoodMart** and **Maa Traders** are the top suppliers, contributing **24.1%** and

23.7% respectively, showing their strong product demand and reliability. **Grocery Depot** also has a significant share of **22.5%**, while **FreshGoods**, **EduWorld**, and **CleanHouse** contribute smaller portions. This distribution shows that the store maintains good relationships with multiple suppliers, ensuring steady stock availability and variety for customers.



The analysis of Vishnu Priya Variety Store’s sales data over the 60-day period from June to August provided meaningful insights into product performance and customer preferences. By maintaining daily records, cleaning and organizing data, and applying statistical and graphical methods, meaningful insights were obtained about sales performance and overall operations. The use of Google Sheets made it possible to identify trends, analyse revenue patterns, and ensure data accuracy. The project emphasizes the importance of using real data for decision-making rather than assumptions, ultimately helping Vishnu Priya Variety Store enhance its operational efficiency and customer satisfaction.